

Dielectric Sphere in a Uniform Electric Field

Source: Cambridge Electromagnetism, Chapter 8 exercises, Question 1; accompanies David Tong's Electromagnetism.

A dielectric sphere of radius a and permittivity ε is placed at rest at the origin in a constant electric field which takes a uniform value $\mathbf{E}_0$ far from the origin. Show that Maxwell's equations with the appropriate boundary conditions are solved by an electric field which is uniform inside the sphere, with value:

$$\mathbf{E}_{\text{int}} = \frac{3}{2 + \varepsilon_r} \mathbf{E}_0, \quad (1.1)$$

where $\varepsilon_r = \varepsilon/\varepsilon_0$, and a field outside the sphere which is the superposition of $\mathbf{E}_0$ and the field

$$\mathbf{E}(\mathbf{x}) = \frac{1}{4\pi\varepsilon_0} \left(\frac{3(\mathbf{p} \cdot \hat{\mathbf{x}})\hat{\mathbf{x}} - \mathbf{p}}{|\mathbf{x}|^3} \right) \quad (1.2)$$

of an electric dipole at the origin, with dipole moment

$$\mathbf{p} = 4\pi\varepsilon_0 \left(\frac{\varepsilon_r - 1}{\varepsilon_r + 2} \right) a^3 \mathbf{E}_0. \quad (1.3)$$

By considering the macroscopic polarisation $\mathbf{P}$ inside the sphere, calculate the bound surface charge density σ . Verify explicitly that the electric dipole moment of σ is $\mathbf{p}$.

Solution. Inside and outside the dielectric, the potential satisfies the Laplace equation,

$$\nabla^2 \phi = 0. \quad (1.4)$$

Since the problem is invariant under rotation about $\mathbf{E}_0$, which we will take to be in the z -direction, the solution should have no ϕ dependence. Then, the general solution for the Laplace equation is

$$\phi(r, \theta) = \sum_{l=0}^{\infty} (A_l r^l + B_l r^{-(l+1)}) P_l(\cos \theta). \quad (1.5)$$

Inside the dielectric, we require regularity at the origin; we set $B_l = 0$. Outside the dielectric, since there is a constant electric field at $\mathbf{E}_0 = E_0 \cos \theta \hat{\mathbf{z}}$ $r \rightarrow \infty$, we take $A_l = -E_0 \delta_{l1}$. So the general solution we are looking at is

$$\phi(r, \theta) = \begin{cases} \sum_{l=0}^{\infty} A_l r^l P_l(\cos \theta), & r < a, \\ \sum_{l=0}^{\infty} D_l r^{-(l+1)} P_l(\cos \theta) - E_0 r \cos \theta, & r > a. \end{cases} \quad (1.6)$$

The boundary condition at $r = a$ is given by

$$\hat{\mathbf{r}} \times (\mathbf{E}(a\hat{\mathbf{r}}^+) - \mathbf{E}(a\hat{\mathbf{r}}^-)) = 0, \quad \text{and} \quad \hat{\mathbf{r}} \cdot (\varepsilon_0 \mathbf{E}(a\hat{\mathbf{r}}^+) - \varepsilon \mathbf{E}(a\hat{\mathbf{r}}^-)) = 0, \quad (1.7)$$

which implies

$$\partial_\theta \phi(a^+, \theta) = \partial_\theta \phi(a^-, \theta), \quad \text{and} \quad \varepsilon_0 \partial_r \phi(a^+, \theta) = \varepsilon \partial_r \phi(a^-, \theta). \quad (1.8)$$

The first BC gives

$$A_l a^l = D_l a^{-(l+1)} - E_0 a \delta_{l1}, \quad (1.9)$$

and the second BC gives

$$\varepsilon_0 l A_l a^{l-1} = -\varepsilon(l+1) D_l a^{-(l+2)} - \varepsilon E_0 \delta_{l1}. \quad (1.10)$$

Solving these two, we find

$$A_l = -\frac{3}{2 + \varepsilon_r} E_0 \delta_{l1} \quad \text{and} \quad D_l = -\frac{1 - \varepsilon_r}{2 + \varepsilon_r} a^3 E_0 \delta_{l1}. \quad (1.11)$$

Hence, the full potential is

$$\phi(r, \theta) = \begin{cases} -\frac{3}{2 + \varepsilon_r} E_0 r \cos \theta, & r < a, \\ -\frac{1 - \varepsilon_r}{2 + \varepsilon_r} \frac{a^3}{r^2} E_0 \cos \theta - E_0 r \cos \theta, & r > a. \end{cases} \quad (1.12)$$

Therefore, the electric field inside the dielectric is

$$\mathbf{E}_{\text{int}} = \frac{3}{2 + \varepsilon_r} E_0 (\cos \theta \hat{\mathbf{r}} - \sin \theta \hat{\theta}) = \frac{3}{2 + \varepsilon_r} E_0 \hat{\mathbf{x}}. \quad (1.13)$$

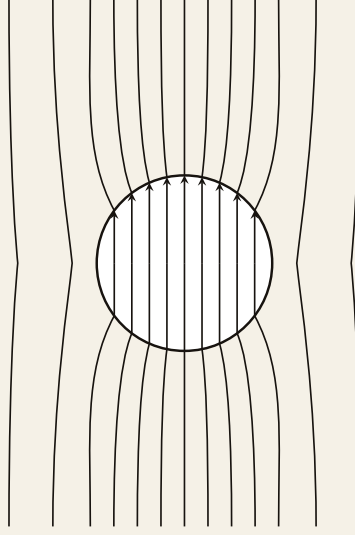
The electric field outside is

$$\mathbf{E}(\mathbf{x}) + \mathbf{E}_0 = \frac{1 - \varepsilon_r}{2 + \varepsilon_r} a^3 \frac{(\mathbf{E}_0 \cdot \hat{\mathbf{r}}) \mathbf{r} - \mathbf{E}_0}{r^3} + \mathbf{E}_0 = \frac{1}{4\pi\varepsilon_0} \left(\frac{3(\mathbf{p} \cdot \hat{\mathbf{x}}) \hat{\mathbf{x}} - \mathbf{p}}{|\mathbf{x}|^3} \right) + \mathbf{E}_0 \quad (1.14)$$

with

$$\mathbf{p} = 4\pi\varepsilon_0 \left(\frac{\varepsilon_r - 1}{\varepsilon_r + 2} \right) a^3 \mathbf{E}_0. \quad (1.15)$$

A sketch of the electric field is shown in Fig. 1.



Electric field lines for a dielectric sphere ($\epsilon_r = 3$, $a = 1$) in a uniform applied field $\mathbf{E}_0$. Inside the field is uniform and weaker, $\mathbf{E}_{\text{int}} = \frac{3}{5}\mathbf{E}_0$; outside it is the superposition of $\mathbf{E}_0$ and an electric dipole. Lines refract at the boundary satisfying continuity of $E_{\parallel}$ and $D_{\perp}$.

The polarisation vector is given by

$$\mathbf{P} = (\epsilon_r - 1)\epsilon_0\mathbf{E}_{\text{int}}. \quad (1.16)$$

The bound charge density is simply given by

$$\sigma(\theta) = \hat{\mathbf{r}} \cdot \mathbf{P} = \frac{3(\epsilon_r - 1)}{2 + \epsilon_r}\epsilon_0 E_0 \cos \theta. \quad (1.17)$$

The dipole due to the surface charge distribution is, therefore, given by

$$\mathbf{p} = \int_{-1}^1 d\cos \theta \, 2\pi\sigma(\theta)a^2 \cos \theta = 4\pi\epsilon_0 \left(\frac{\epsilon_r - 1}{\epsilon_r + 2} \right) a^3 \mathbf{E}_0, \quad (1.18)$$

which indeed matches what we obtained by solving the Laplace equation.